1. Introduction

Women's safety in public spaces is an important concern, and effective reporting and monitoring of safety incidents can help authorities take appropriate preventive measures. Conventional methods of reporting incidents may involve delays and may not provide a centralized platform for authorities to monitor incidents based on location, incident type, date, time and risk level.

The proposed project develops a cross-platform Safety Mobile Application for iOS and Android, along with an Authority Dashboard, as the user interaction and monitoring component of the system. The mobile application allows users to report women-safety incidents along with relevant information such as incident type, date, time, description and geographical location. The reported information is transmitted to the backend through APIs and stored in the database.

The authority dashboard provides a centralized interface for authorized personnel to monitor reported incidents and the analytical results generated by the incident analysis module. It displays incident statistics, geographical information, identified hotspots and risk-level information through maps, charts and other visualizations. Authorities can filter the information based on geographical region or city, incident type, date and time, and risk level.

The system also provides automated daily alerts based on recent incident history. These alerts can highlight locations where incidents have occurred or repeatedly occurred and can support authorities in planning preventive patrolling and safety measures.

The application and dashboard are designed to support different geographical regions and cities based on the availability of relevant incident and geographical data. The currently available dataset contains Bengaluru-specific geographical and infrastructure information, while the system architecture is designed to accommodate other regions and cities.

2. Literature Survey

2.1 Mobile Applications for Public Safety

Mobile applications have become an accessible platform for users to report safety-related incidents. Such applications provide convenient interfaces for submitting incident information and can reduce the difficulties associated with conventional reporting methods. Mobile-based reporting can also allow incident information to be collected along with geographical details.

2.2 Location-Based Incident Reporting

Location-based reporting associates an incident with its geographical coordinates. Capturing location information along with incident details helps authorities understand where incidents occur and supports geographical analysis and visualization.

2.3 Authority Dashboards

Dashboards provide a centralized interface for monitoring and managing large amounts of information. In public-safety systems, dashboards can display incident statistics, locations, trends and other relevant information in an organized manner for authorized personnel.

2.4 Interactive Maps for Safety Monitoring

Interactive maps provide a visual representation of incident locations and identified hotspots. Map-based visualization enables authorities to understand the geographical distribution of incidents and identify areas that require greater attention.

2.5 Filtering and Data Exploration

Filtering mechanisms allow authorities to examine specific portions of incident data. Filters based on geographical region, incident type, date, time and risk level can help authorities focus on relevant information and understand incident patterns within selected areas.

2.6 Automated Alerts

Automated alerts provide timely information about recently reported incidents and locations with recurring incidents. Such alerts can help authorities identify areas requiring attention and support preventive safety actions.

2.7 API-Based System Integration

Application Programming Interfaces (APIs) enable communication between the mobile application, backend services, database and analytical components. API-based integration allows newly submitted incident information to be transferred between system components and made available for analysis and visualization.

2.8 Research Gap

Existing safety applications and dashboards often focus on individual functions such as incident reporting, visualization or monitoring. There is a need for an integrated system that connects mobile incident reporting with centralized authority monitoring, geographical visualization, filtering, analytical results and automated alerts. The proposed system addresses this gap by connecting the mobile application with the backend, database and incident-analysis component to provide updated safety information to authorities.

3. Problem Statement, Aim and Objectives

3.1 Problem Statement

Women-safety incidents in public spaces need to be reported and monitored effectively so that authorities can identify locations requiring greater attention. Conventional reporting methods may not provide a centralized platform for collecting location-based incident information and monitoring it through maps, statistics and filters.

There is a need for a system that allows users to report incidents through a mobile application and enables authorities to monitor the collected information through a centralized dashboard. The system should provide location-based visualization, filtering based on relevant parameters, analytical results such as hotspots and risk levels, and automated alerts based on recent incident history to support preventive patrolling.

3.2 Aim

To develop a cross-platform Safety Mobile Application for iOS and Android and an Authority Dashboard that enable location-based incident reporting, centralized monitoring, filtering and visualization of analytical results, and automated alerts to support preventive safety measures.

3.3 Objectives

- To develop a cross-platform mobile application for iOS and Android for reporting women-safety incidents.
- To capture incident information such as incident category, location, date, time and description.
- To transmit incident information from the mobile application to the backend through APIs.
- To store and manage submitted incident information through a centralized backend and database.
- To provide an authority dashboard for monitoring incident statistics, geographical information, hotspots and risk levels.
- To provide filtering based on geographical region or city, incident type, date and time, and risk level.
- To visualize incident locations and analytical results using interactive maps and charts.
- To provide automated daily alerts based on recent incident history and location-based information to support preventive patrolling.
- To provide authentication and access control for authorized dashboard users.

- To design the application and dashboard to support different geographical regions and cities based on available data.

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4. SDG & ESG Compliance

A. Section A — SDG Compliance

i. Project title: Safety Mobile Application and Authority Dashboard

ii. Identified SDG (Goal No. & Name):

Sustainable Development Goal (SDG): SDG 5 – Gender Equality

iii. Target & Indicator under this SDG:

Target 5.2 – Eliminate all forms of violence against all women and girls in public and private spheres, including trafficking, sexual exploitation, and other forms of violence.

Indicator 5.2.2 – Proportion of women and girls subjected to sexual violence by persons other than an intimate partner, by age and place of occurrence.

iv. Contribution of the Project towards the identified SDG:

This project aligns with SDG 5 by addressing women's safety in public spaces and supporting preventive measures against harassment. Its contribution can be summarised as follows:

- The system analyses women-safety incidents and identifies locations where incidents are concentrated or repeatedly occurring.
- Hotspot visualization and automated alerts give authorities information that can support preventive patrolling and timely intervention.
- The mobile application delivers safety-oriented functionality to users, contributing to safer participation in public spaces.

v. Supporting Documents / Evidence:

- Project synopsis describing the objectives, methodology and implementation of the Safety Mobile Application and Authority Dashboard, along with its alignment with SDG 5 – Gender Equality.
- Screenshots of the safety mobile application demonstrating incident reporting, location capture and other safety-oriented features for users.
- Sample incident reports submitted through the mobile application, including incident type, location, date, time and description.
- Sample API request and response outputs demonstrating communication between the mobile application, backend and database.
- Screenshots of the authority dashboard displaying incident information, statistics, geographical locations, hotspots and risk-level information.
- Screenshots demonstrating filtering and region-based analysis based on geographical region or city, incident type, date and time, and risk level.
- Sample automated daily alerts generated from recent incident history to highlight locations requiring increased attention.
- Sample dashboard outputs demonstrating how location-based incident information and alerts can support preventive patrolling and safety planning.

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B. Section B — ESG Compliance

ESG Category	Project Contribution	Supporting Documents
Environmental (E)	The system uses digital reporting and storage instead of depending on paper records. This helps reduce paper usage and makes incident records easier to maintain digitally.	System design, database design, web/mobile application screenshots
Social (S)	The system helps authorities and users understand where and when women's safety incidents occur more frequently. By identifying recurring incidents and high-risk areas, it supports better patrolling and safety planning. Interactive maps and dashboards make the information easier to understand and help authorities make more informed decisions based on actual incident data.	Project synopsis, hotspot maps, authority dashboard, mobile application screenshots, SDG alignment
Governance (G)	The system protects incident information by using authentication and role-based access. This ensures that only authorized users can access the data and helps maintain the privacy and security of sensitive information.	System architecture, authentication module, access-control design, security documentation

Signature of Student **Signature of Project Guide**

5. Methodology

The proposed system follows an integrated workflow connecting the safety mobile application, backend, database, analytical component and authority dashboard.

System Flow

User → Safety Mobile Application → Incident Report → API → Backend → Database → Module 1 Analysis → Analytical Results → Authority Dashboard → Filtering → Alerts & Patrolling Support

5.1 Safety Mobile Application

A cross-platform Safety Mobile Application will be developed for both iOS and Android platforms to provide users with an interface for reporting women-safety incidents. The application will collect relevant incident information, capture geographical location and provide basic safety-oriented functionality.

5.2 Location Capture

The application will capture the geographical location associated with an incident using the mobile device's location services. Latitude and longitude information will be associated with the incident record and used for geographical analysis and visualization.

5.3 Incident Submission

The user will provide information such as incident type, date, time and description and submit the report through the application. The incident information will be transferred to the backend through an API.

5.4 Backend and Database Integration

The backend will receive incident information from the mobile application and store the records in the database. It will also communicate with the analytical component and dashboard so that updated information can be made available to authorized personnel.

5.5 Authority Dashboard

A centralized dashboard will be provided for authorized personnel. The dashboard will display reported incidents, incident statistics, geographical information, identified hotspots and risk-level information generated by the analytical component.

5.6 Interactive Map Visualization

Incident locations and identified hotspots will be represented using interactive maps. Charts and statistical visualizations will also be provided to help authorities understand incident distribution, frequency and geographical concentration.

5.7 Filtering and Region-Based Analysis

The dashboard will allow authorities to select a geographical region or city and filter information based on incident type, date and time, and risk level. This will allow authorities to focus on specific areas and relevant incident records.

5.8 Automated Alerts and Patrolling Support

The system will generate daily alerts based on recent incident history. These alerts can highlight locations where incidents have occurred or repeatedly occurred and provide location-based information to support preventive patrolling.

5.9 Continuous Data Update

Whenever a new incident is submitted through the mobile application, the information will be stored in the backend and database. The analytical component can incorporate the new record and update relevant incident statistics, hotspot information and risk-level information displayed on the dashboard.

5.10 Authentication and Access Control

Authentication and access control will be implemented to ensure that only authorized personnel can access the authority dashboard and administrative information.

6. Modules Identified

6.1 User Incident Reporting

This module provides the interface for users to report women-safety incidents. Users can provide details such as incident category, date, time, location and description.

6.2 Location Capture

This module captures the geographical location associated with an incident using the mobile device's location services. The captured coordinates are used for location-based analysis and visualization.

6.3 Incident Submission & API Communication

This module transfers incident information from the mobile application to the backend through APIs. It ensures that submitted incident details are communicated in a structured format.

6.4 Backend & Database Integration

This module manages communication between the mobile application, backend, database and analytical component. Submitted incident records are stored in the database and made available for analysis.

6.5 Authority Dashboard

This module provides a centralized interface for authorized personnel to monitor incident information. It displays incident statistics, hotspots and risk-level information obtained from the analytical component.

6.6 Map & Data Visualization

This module represents incident locations, hotspots and statistical information through interactive maps, charts and other visualizations to make the information easier to understand.

6.7 Filtering & Region-Based Analysis

This module allows authorities to select a geographical region or city and filter incident information based on incident type, date and time, and risk level. It helps authorities focus on specific areas and relevant incident records.

6.8 Automated Alerts & Patrolling Support

This module provides daily alerts based on recent incident history and location-based information. It highlights areas with incidents or recurring incidents and supports authorities in planning preventive patrolling.

7. Tools and Technologies Identified

Table No: 7.1 Tools and technologies

Category	**Tools / Technologies**
Mobile Application	Flutter / React Native – iOS and Android
Backend	Python, FastAPI
Database	MongoDB
Data Processing	Pandas, NumPy
Geospatial Processing	GeoPandas
Maps	OpenStreetMap, Leaflet/Folium
Visualization	Matplotlib, Plotly
Dashboard / Frontend	HTML, CSS, JavaScript, React
API Communication	REST API, JSON
API Testing	Postman
Development Environment	Visual Studio Code, Android Studio
Version Control	Git and GitHub
Data Source	Existing Excel Dataset, Mobile Incident Data

8. Timeline using Gantt Chart

Table No: 8.1 Gantt Chart

Activities	W1-W2	W3-W4	W5-W6	W7-W8	W9-W10	W11-W12	W13-W14	W15-W16
Problem Identification & Requirement Analysis	✓	✓						
Literature Survey	✓	✓	✓	✓				
Mobile Application Design		✓	✓					
User Incident Reporting		✓	✓	✓				
Location Capture		✓	✓	✓				
Backend & Database Integration			✓	✓	✓			
API Communication				✓	✓	✓		
Authority Dashboard				✓	✓	✓	✓	
Map & Data Visualization				✓	✓	✓	✓	
Filtering & Region-Based Analysis					✓	✓	✓	
Automated Alerts & Patrolling Support						✓	✓	
Authentication & Access Control					✓	✓		
Integration with Analytical Module					✓	✓	✓	
Testing & Validation						✓	✓	✓
Documentation & Demonstration							✓	✓

Expected Deliverable

At the end of the implementation, the project is expected to deliver a functional Safety Mobile Application and Authority Dashboard integrated with the backend, database and analytical component.

The mobile application will allow users to submit women-safety incident reports along with relevant details such as incident type, date, time, description and geographical location. The submitted information will be transmitted through APIs and stored in the backend database.

The Authority Dashboard will provide a centralized interface for authorized personnel to monitor reported incidents and analytical results. It will display incident statistics, geographical information, identified hotspots and risk-level information through interactive maps and visualizations.

The dashboard will support filtering based on geographical region or city, incident type, date and time, and risk level, allowing authorities to focus on specific locations and incident categories.

The system will also provide automated daily alerts based on recent incident history and location-based information. These alerts will help identify areas with concentrated or recurring incidents and support authorities in planning preventive patrolling and safety measures.

Whenever new incident reports are submitted through the mobile application, the information will be incorporated into the backend and made available to the analytical component so that relevant incident statistics, hotspot information and risk-level information can be updated.

The system will be designed to support different geographical regions and cities based on the availability of relevant data. The currently available Excel dataset will be used for Bengaluru, while the overall analytical and application workflow will remain applicable to other regions.

Final Deliverables

- Functional cross-platform Safety Mobile Application for iOS and Android
- Backend and database integration
- Authority Dashboard
- Interactive maps and data visualizations
- Region and incident-based filtering
- Hotspot and risk-level information from the analytical module
- Automated daily alerts
- Location-based preventive patrolling support
- Authentication and access control
- API integration between the mobile application, backend and analytical component
- Testing and validation results